# Specific-factors model with immigration

Two sectors : $M$ & $A$

$Q_M = F_M(K, L_M)$

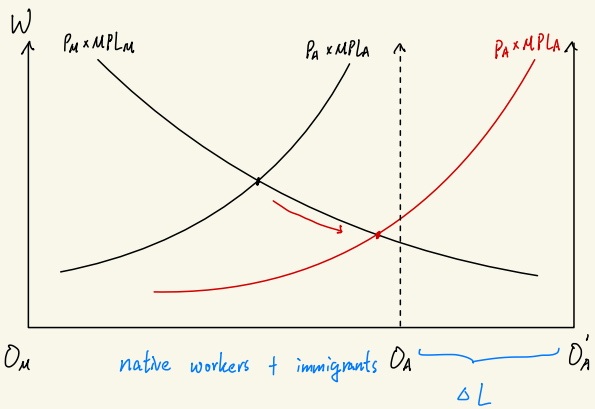
$Q_A = F_A(T, L_A)$

$L_M + L_A = L$

$W = P_M \, MPL_M = P_A \, MPL_A$

$r_K = P_M \, MPK$ , $r_T = P_A \, MPT$

## 1. More mobile workers




$W \downarrow$

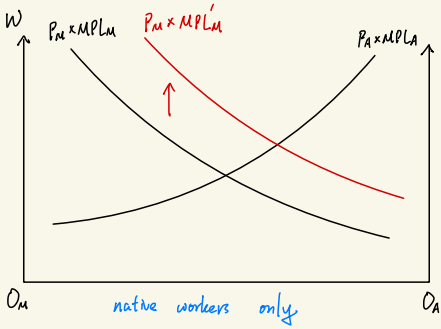
$L_M \uparrow \Rightarrow MPK \uparrow \Rightarrow r_K \uparrow$

$L_A \uparrow \Rightarrow MPT \uparrow \Rightarrow r_T \uparrow$

## 2. Immigrants are M-specific but complement with native workers

$Q_M = F(K, I, L) \Rightarrow MPL_M(K, I, L)$ (+) (+) (-)

e.g. $Q_M = K^{1-\alpha-\beta} I^{\beta} L^{\alpha} \Rightarrow MPL = \alpha K^{\overset{1-\alpha-\beta}{\downarrow 0}} I^{\underset{\downarrow 0}{\beta}} L^{\overset{\alpha-1}{\uparrow 0}}$

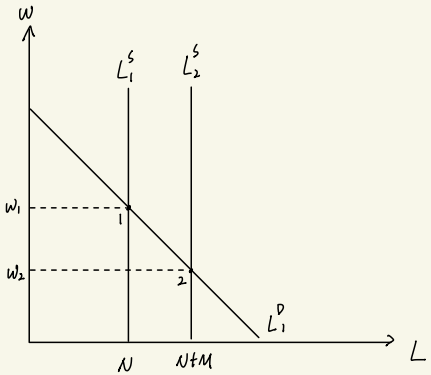



$I \uparrow \Rightarrow MPL_M \uparrow \Rightarrow W \uparrow \Rightarrow L_M \uparrow, L_A \downarrow$

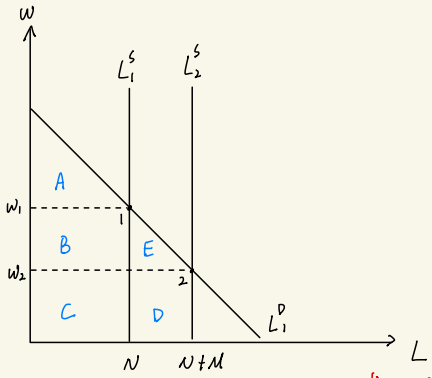
• Immigration and Capital adjustment (NAS, 2017)

One sector : $Y = K^{\alpha} L^{1-\alpha}$ $(A = 1)$

$w = (1-\alpha)\left(\frac{K}{L}\right)^{\alpha}$ , $r = \alpha\left(\frac{K}{L}\right)^{\alpha-1}$ , $Y = wL + rK$




$L_1^S \to L_2^S \Rightarrow \frac{K}{L} \downarrow \Rightarrow w \downarrow$ . $r \uparrow$




native immigrant

$Y_1 = (B + C) + A$ , $Y_2 = (C + D) + (A + B + E)$

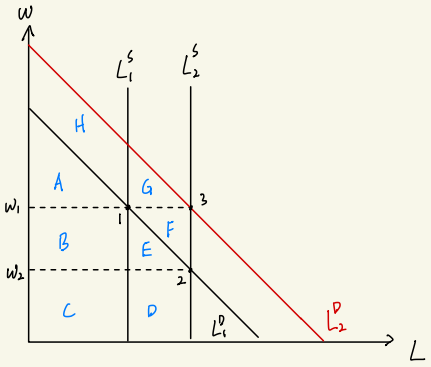
$wL_1$ $rK_1$ $wL_2$ $rK_1$

$\Rightarrow \Delta Y = Y_2 - Y_1 = D + E$

immigrant income, native income

In the short run, $r \uparrow \Rightarrow$ capital inflows $\Rightarrow K \uparrow$

In the long run, $r = r^*$




$K \uparrow \Rightarrow MPL \uparrow \Rightarrow w \uparrow$

$Y_3 = (B + C + D + E + F) + (A + G + H)$

native income, immigrant income, origin capital, additional capital

# • Perfect Substitutes and Perfect Complements

→ Perfect substitutes : two inputs can replace each other at a constant rate

① Without capital

$Q = N + M$

⇒ $MPL_N = \frac{\partial Q}{\partial N} = 1$ is unchanged when $M \uparrow$

② With capital

$Q = K^{\alpha}(N+M)^{(1-\alpha)}$ $(<1)$

⇒ $MPL_N = (1-\alpha)\left(\frac{K}{N+M}\right)^{\alpha}$. so $M \uparrow \Rightarrow MPL_N \downarrow$

→ Complements : one input makes other inputs more productive

$Q = F(N, M)$. When $M \uparrow$, $MPL_N \uparrow$

e.g. $Q = N^{\alpha} M^{1-\alpha}$

$MPL_N = \alpha\left(\frac{M}{N}\right)^{1-\alpha}$. so $M \uparrow \Rightarrow MPL_N \uparrow$

→ Perfect Complements : inputs must be used together in fixed proportions

$Q = \min\{N, M\}$

Suppose $N = 10$

$M = 5 \Rightarrow Q = 5$

$M = 10 \Rightarrow Q = 10$

$M = 15 \Rightarrow Q = 10$